Team Persistent or Team Transitory?

Sectoral Linkage and Inflation Persistence

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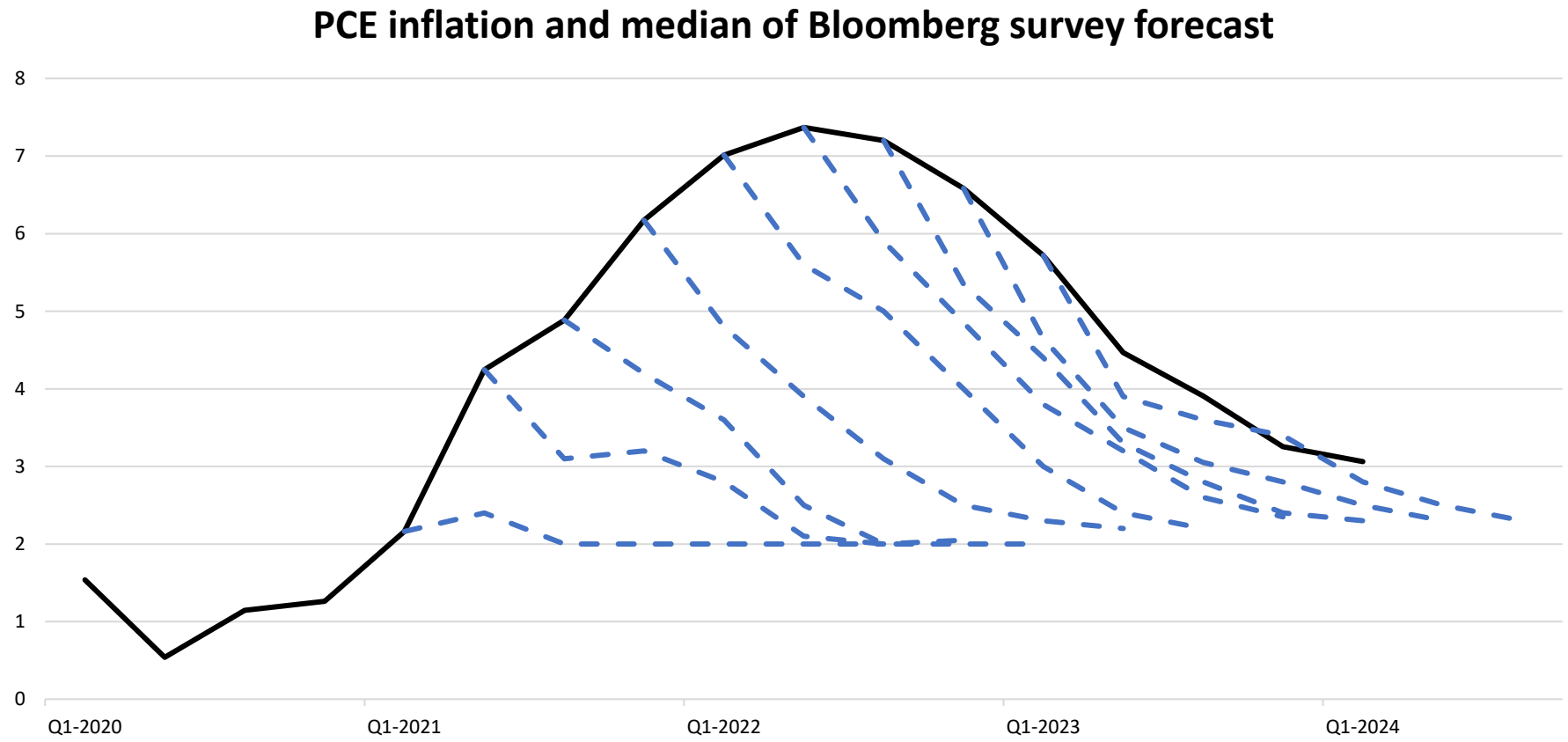
Network conference @
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Motivation

- US (and world) inflation has been high since 2021

Motivation

- US (and world) inflation has been high since 2021
- It is also much more persistence than many people think



- Both the market and the Fed have it wrong

Research question

- Why is inflation so stubborn this time?

A LOT of sectoral shocks….

- Why is inflation so stubborn this time?

"All the News That's Fit to Print"

The New York Times

Late Edition

Today, periodic clouds and sunshine, a bit mild, high 43. Tonight, partly to mostly cloudy, low 35. Tomorrow, clouds and sunshine, windy, high 41. Weather map appears on Page A20.

VOL. CLXXI .... No. 59,302 © 2022 The New York Times Company NEW YORK, THURSDAY, JANUARY 13, 2022 $3.00

Chip Shortage Dents Passenger Vehicle Sales in Nov

PV makers say global shortage hit production while two-wheeler makers blame rise in acquisition cost … on to Bharat-VI

Our Bureau

New Delhi: Domestic sales of both cars and two-wheelers fell last month though due to comple-ely different reasons even as low demand from small transporters impacted recovery in the commer-cial vehicle segment, industry of-ficials said on Wednesday.

Passenger vehicle makers said they could not meet a strong con-sumer demand as a global shorta-ge of semiconductors continued to impact production while two-wh-eler makers blamed rise in acquisition costs after … rat Stage-VI along with e… second wave … areas for a slu… Industry est… 246,000 passen… sold in the local …

down 14% from an estimated 286,800 units sold in November 2020.

Automakers in India mostly re-port wholesale dispatches from factories and not retail sales by de-alers to customers.

Market leader Maruti Suzuki's wholesale volumes in the domestic market declined 19% year on year to 109,736 units last month.

"While demand remains intact, the shortage of semicond… have been affecte… operations … rers."

Core Issue PV Sales (Domestic Only)

Company	Nov-20	Nov-21	% chg
Maruti Suzuki	135,775	109,…	
Hyundai Motor India	4…		
Tata Motors			
M…			

… the ongoing semi… … shortage situation … that it will continue … forts to miti… …

How… Tata Mo… Mahindra & Mahindra … fared better, reporting an … segment

… auto major Toyota, … operates in India through a joint venture with Kirloskar Group, saw sales rise 53% year on year to 13,003 units during the month under review, partly hel-ped by recent product refresh-

… including a limited … strong, which is duly re-… our booking orders and … our best to cater to … said V Wiselink … general … at … leader Ho-… domestic sales slum-… on year to 226,962 units … November while TVS Motor Company's sales fell 29% to 175,940 units and Bajaj Auto reported a drop of 21% at 144,963 units.

"Delay in harvesting due to the late withdrawal of monsoon in many parts of the country impac-ted the demand post-festive sea-son," Hero MotoCorp said in a statement.

For full report, go to www.economictimes.com

THE WALL STREET JOURNAL.

HEALTH | HEALTHCARE

Health Inflation's Big Hike This Year, in Charts

INFLATION HITS 7%, HIGHEST SINCE 1982; SUPPLY CHAINS LAG

As Data Shows Broadening Price Gains, Fed and Biden Face a Challenge

By JEANNA SMIALEK and ANA SWANSON

… climbed to its highest … 40 years at the end of 2021, … troubling development for Presi-dent Biden and economic policy-makers as rapid price gains erode consumer confidence and cast a shadow of uncertainty over the economy's future.

The Consumer Price Index rose 7 percent in the year through De-cember, and 5.5 percent after stripping out volatile prices such as food and fuel. The last time the main inflation index eclipsed 7 percent was 1982.

Policymakers have spent months waiting for inflation to … pace, and restaurant meals are more expensive, possibly a sign that recent wage increases are be-ginning to contribute to higher prices as employers look to cover higher labor costs.

That price increases are becom-ing more widespread — and creeping into areas that are not so directly affected by the pandemic — is a worrisome development for economic policymakers, who are now poised to respond. Federal Reserve officials have indicated



This paper

- We empirically investigate the source of aggregate US inflation from 1960s (aggregate vs sectoral) and its implications to inflation persistence

- In principle, aggregate inflation can be persistent through

a) Aggregate shock is persistent

b) Shocks to some important sector are persistent (Gabaix 2011 Granularity)

c) Shocks are propagated through latent sectoral linkage
(e.g. but not limited to production network)

- We simultaneously estimate these channels through a factor augmented vector autoregressive model (FAVAR) of US PCE sector inflation rates
(Miao, Phillips, Su 2023)

Results 1: Nature of shocks

- Sectoral shocks:

1) **Very rich and nontrivial sectoral spillover**
2) Inflation persistence through spillover is higher than self persistence for 70% of the sectors
3) Shocks to some sectors are more important
   (a. large weight, b. rich spillover c. self persistence)

e.g. Housing = weight + self persistence
Healthcare = weight + rich spillover

- Aggregate shock is transitory and highly related to oil prices

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- Aggregate shock is transitory and highly related to oil prices

- Implication:

aggregate shock → relatively transitionary inflation

sectoral shock → relatively persistent inflation

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Results 2: Contribution of shocks

- Aggregate shock is the major driver of shocks during low inflation period (90%)

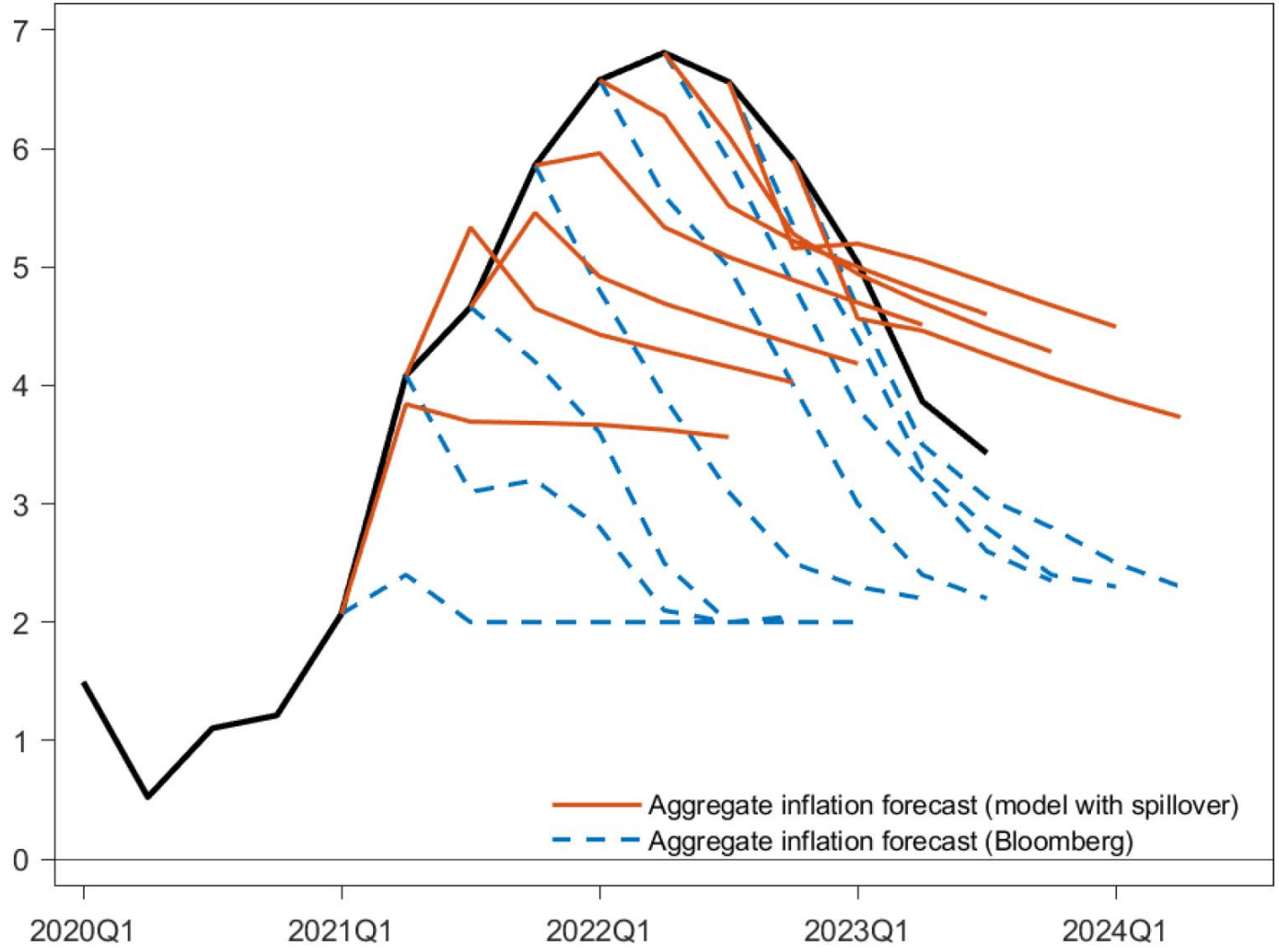
- Sectoral shock contribution increases (45%) during the 2021 inflation episode

- Implication:

Post 2020s inflation is expected to be more persistent

because of large sectoral shocks

Our model prediction



- a much more persistent inflation predicted during 2021-2023
- a hump shape prediction till the peak of inflation at mid 2022

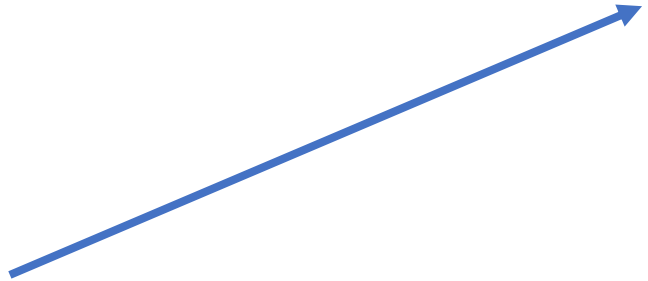
Roadmap

- **Econometric setup**
- Sector shock
- Aggregate shock
- Accounting for post 2020 inflation

Econometric setup

- Based on Miao, Phillips, Su 2023
- A LASSO FAVAR model

$$Y_t = A_1 Y_{t-1} + \Lambda f_t + u_t$$

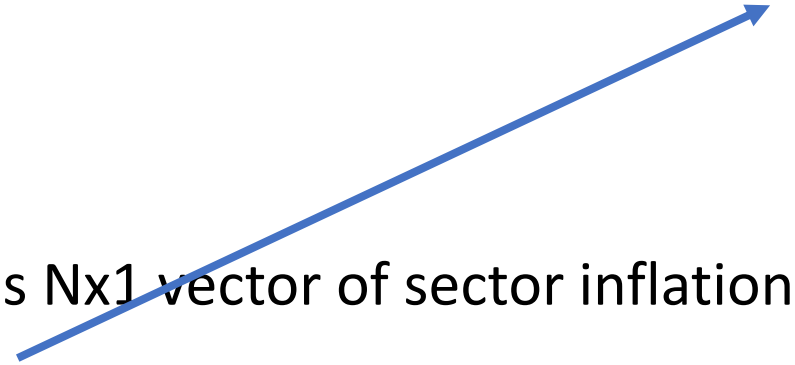


- Y_t is Nx1 vector of sector inflations

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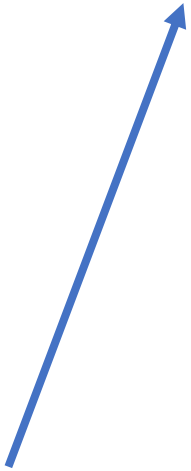
- Y_t is Nx1 vector of sector inflations
- A_1 is the autocorrelation matrix

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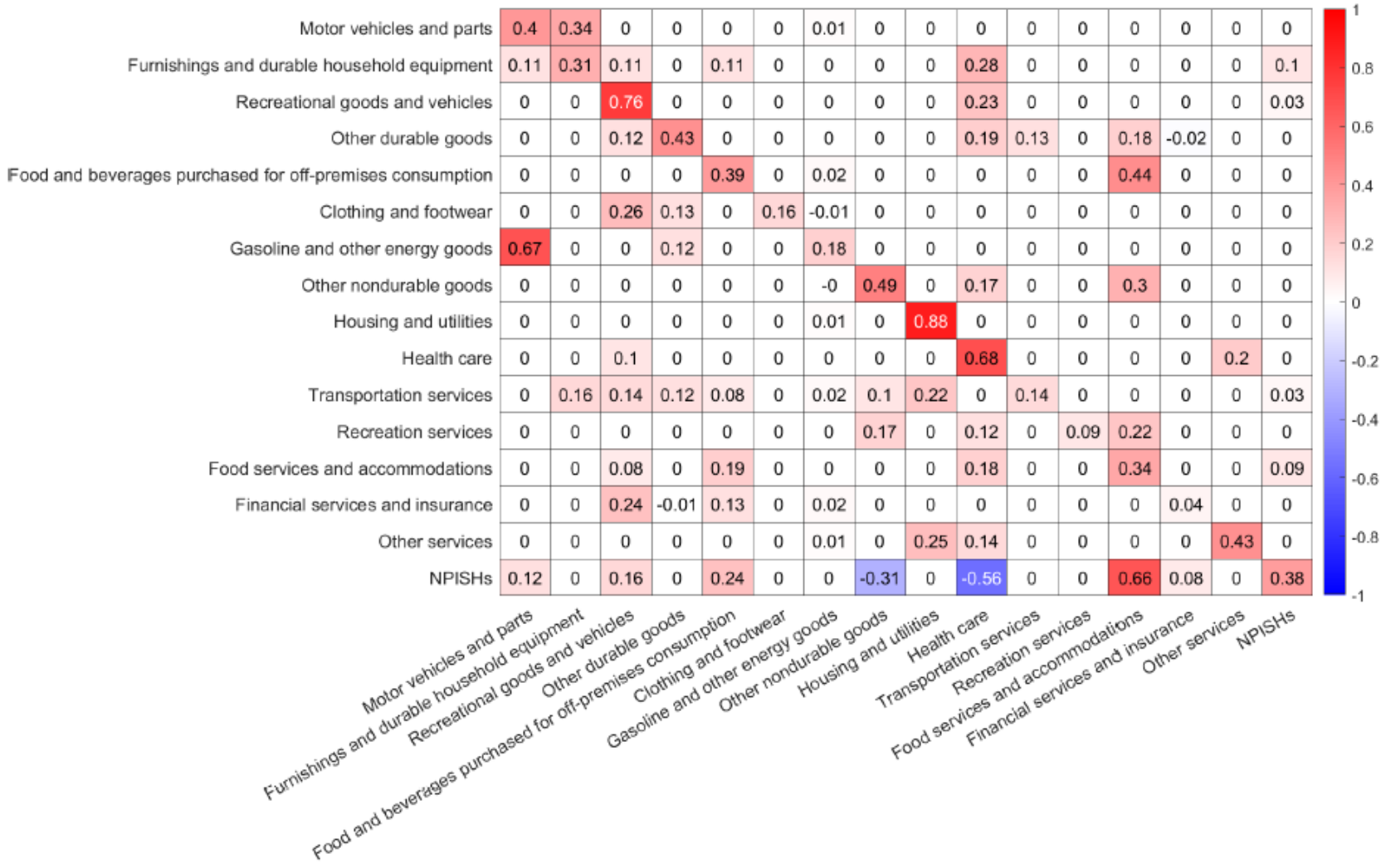


- Y_t is Nx1 vector of sector inflations
- A_1 is the autocorrelation matrix
- f_t is Rx1 vector of common factors, Λ is NxR loading on the common factors

Data

- Data: PCE price index from Bureau of Economic Analysis
- Data range: 1959Q1 - 2023Q3
- Sectors: 16 sectors
- Inflation rates = $100 \times [\log(PCE)_t - \log(PCE)_{t-1}]$

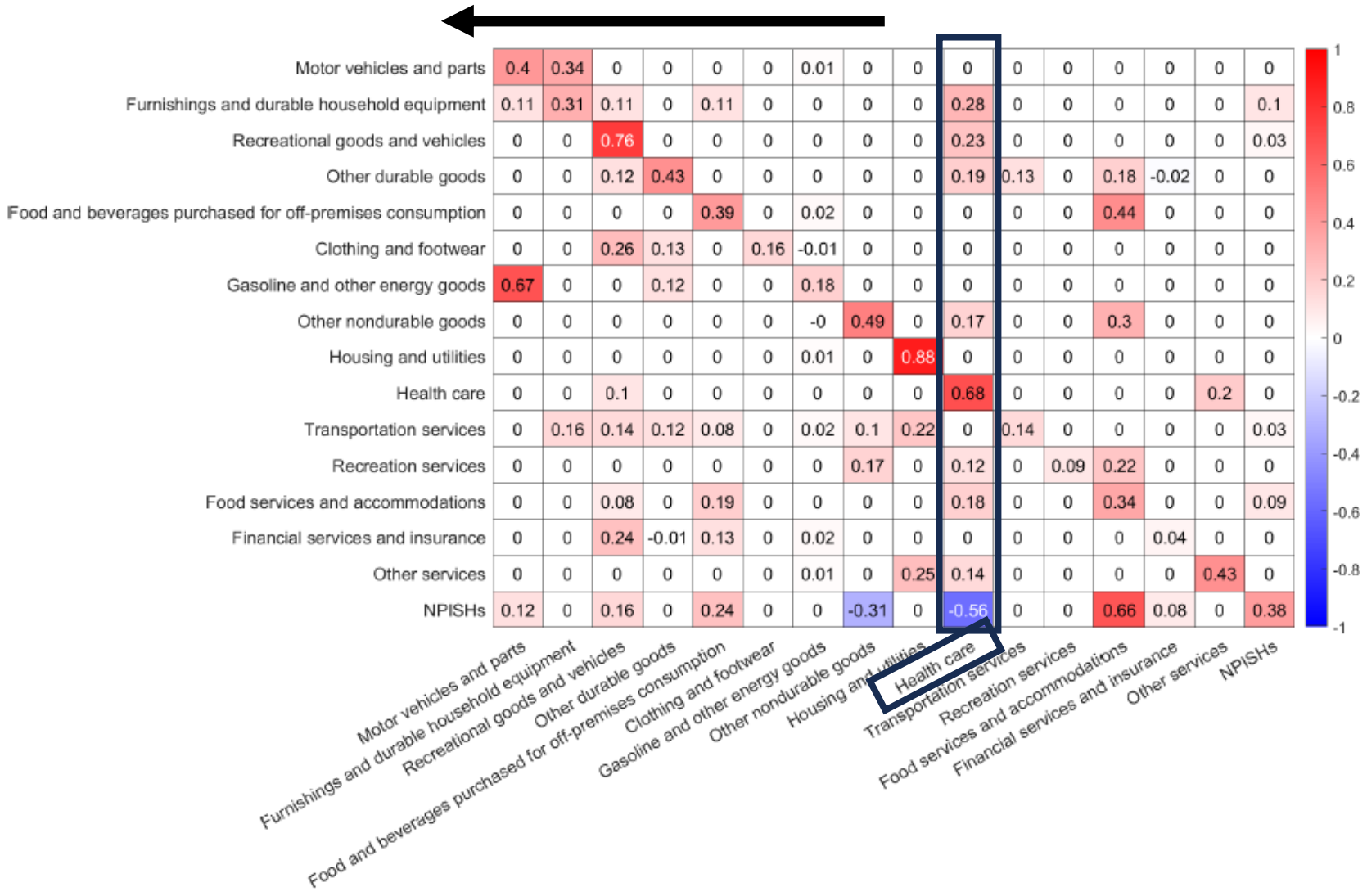
Estimated $\widehat{A}_1$



A column =

how the column sector t-1 inflation is associated with sector inflation in the rows at t

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Key findings of $\widehat{A}_1$

- The diagonal indicates heterogeneous sectoral inflation persistence

- Significant inter-sectoral spillover

30% of off diagonal elements are non-zero (and positive).

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Aggregate IRFs decomposition

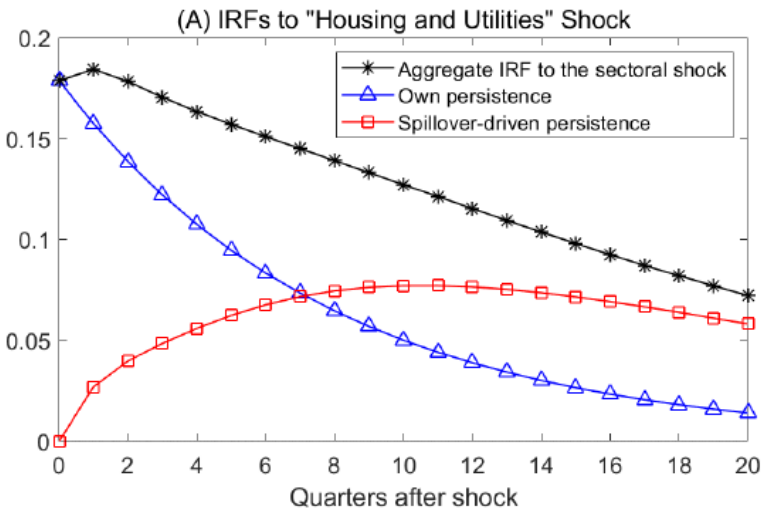
- Decompose the sector IRF of aggregate inflation into:

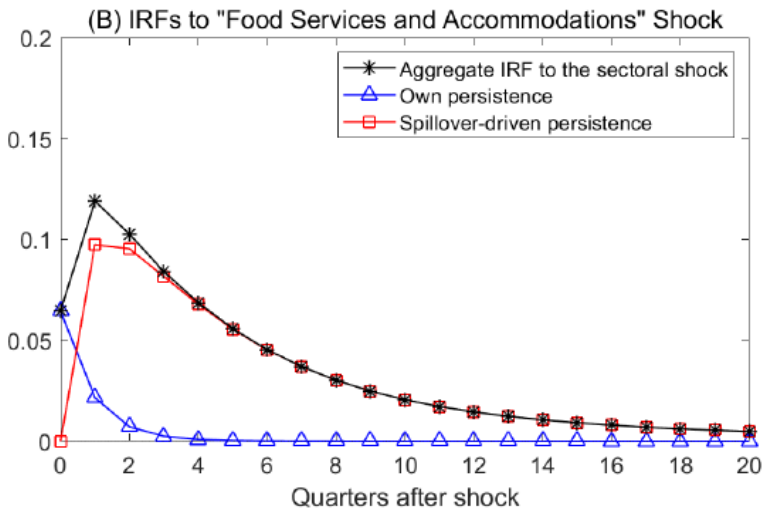
$$IRF^{sector\ j}(h) = \boldsymbol{\omega^T}(\boldsymbol{A_1})_{\boldsymbol{j}}^{\boldsymbol{h}}$$

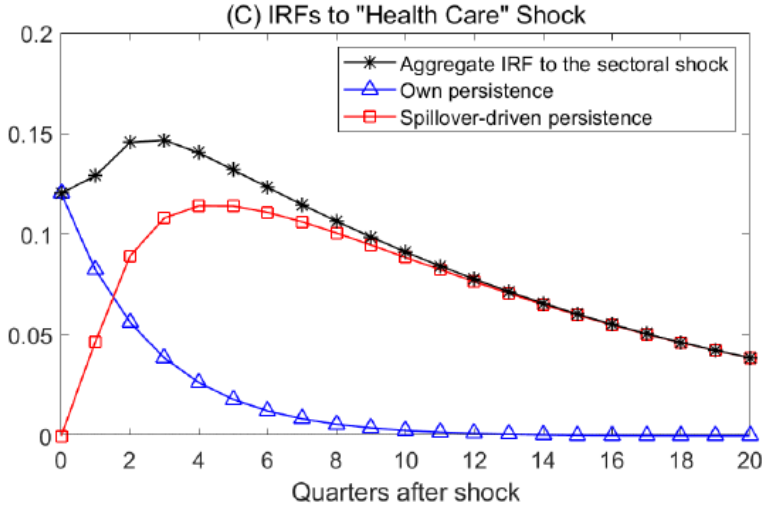
$$= \underbrace{\omega_j((A_1)_{jj})^h}_{\text{Own persistence}} \underbrace{+ \boldsymbol{\omega^T}(\boldsymbol{A_1})_{\boldsymbol{j}}^{\boldsymbol{h}} - \omega_j((A_1)_{jj})^h}_{\text{Spillover persistence}}$$

where $\boldsymbol{\omega^T}$ is a vector of the BEA sector weights

Three major types of sector inflation

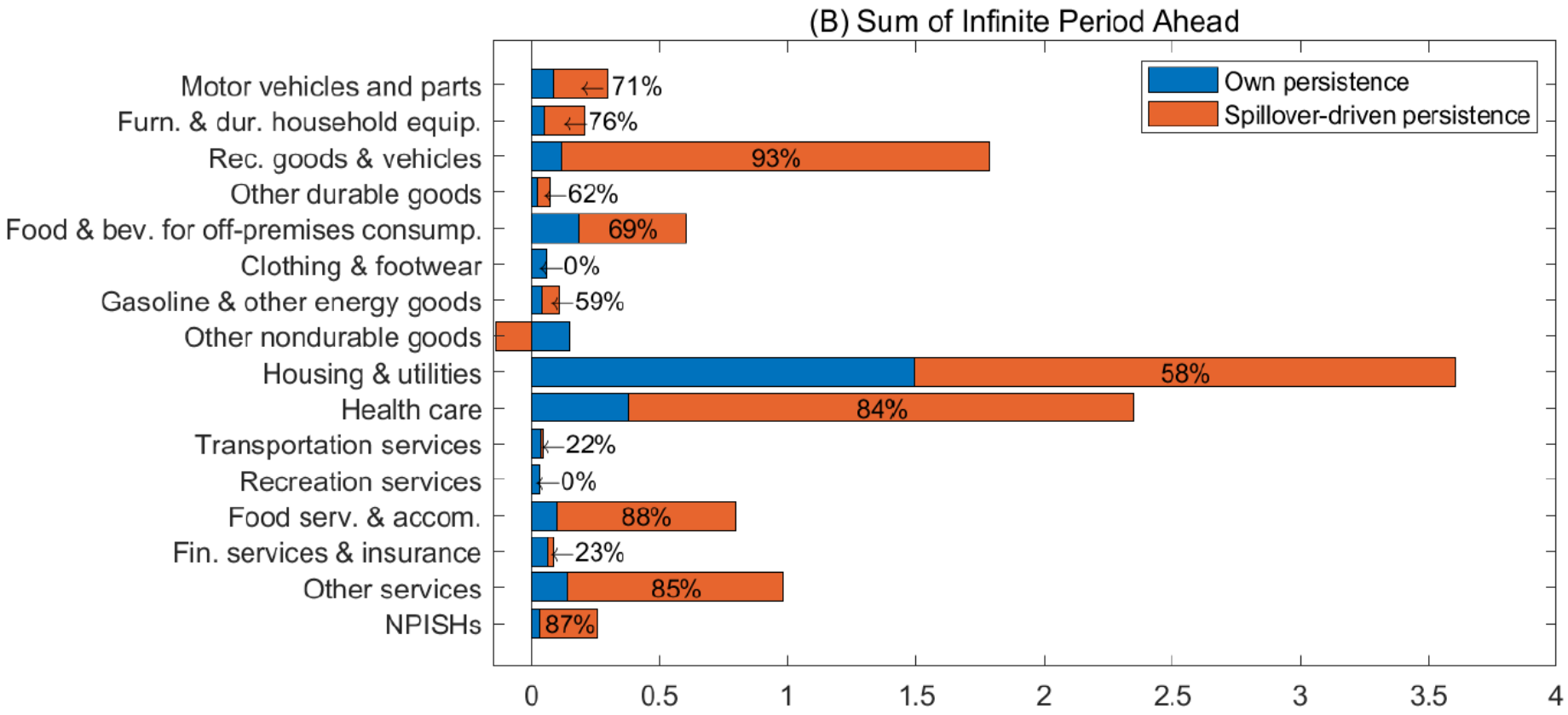






- Type A (Housing): heavy weight + self persistence
- Type B (Food services): heavy weight + rich spillover
- Type C (Healthcare): heavy weight + self persistence + rich spillover
- Hump shape response is possible, half-life of these figures are 10 quarters +

Cumulative inflation (integral under the IRF curve)



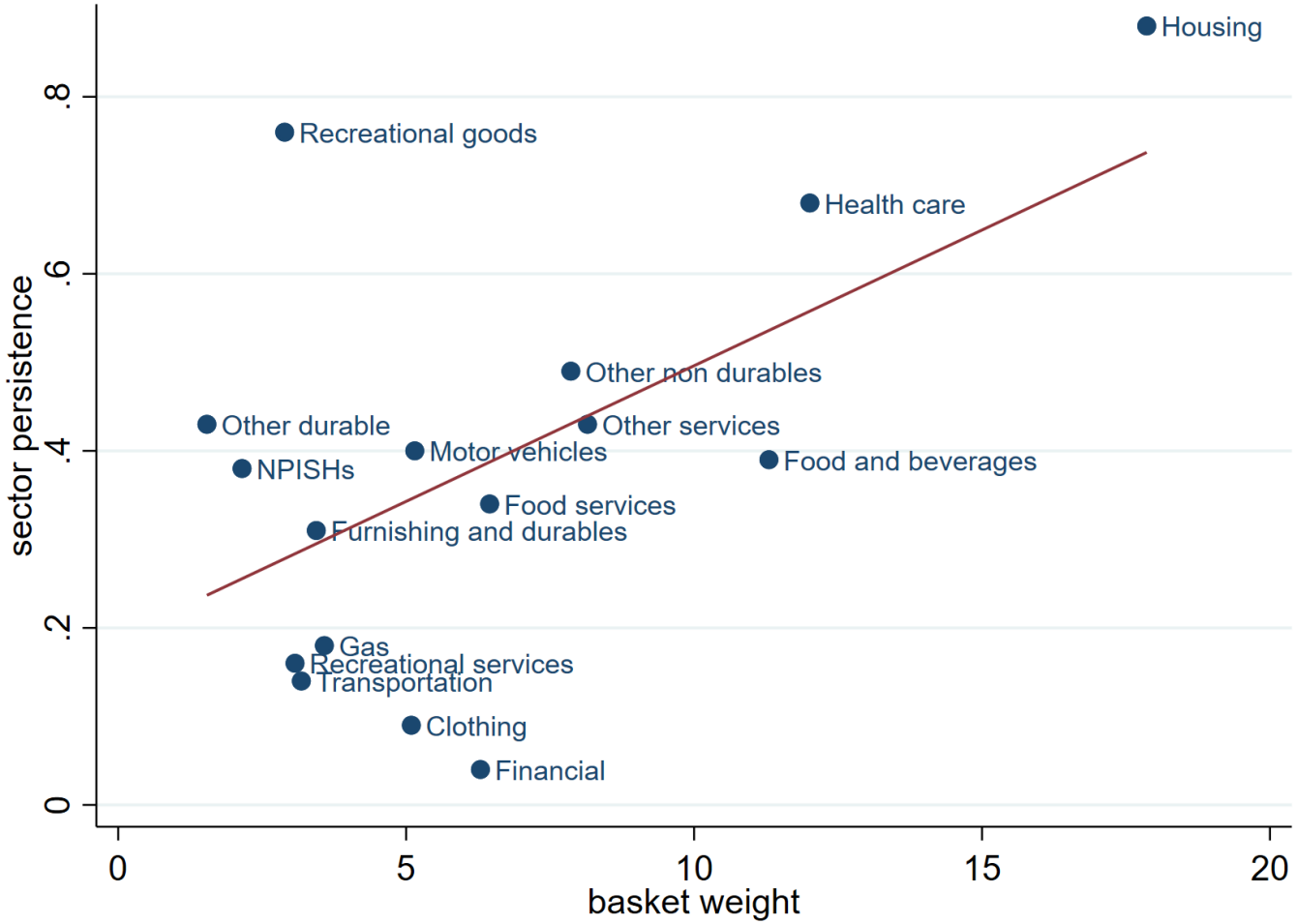
- Absolute cumulative inflation depends a lot on consumption basket weight
- 11 out of 16 sector has spillover inflation > own inflation persistence

Summary of inflation spillover

- Very rich spillover!

- Inflation could be persistence because of rich spillover

Sector granularity or heterogeneity



- Heavy weight sector has a high self persistence (diagonal element of matrix)
- Note: no mechanical relationship between the two

Sector granularity

- Give an uniform sector shock, for each sector i, we have

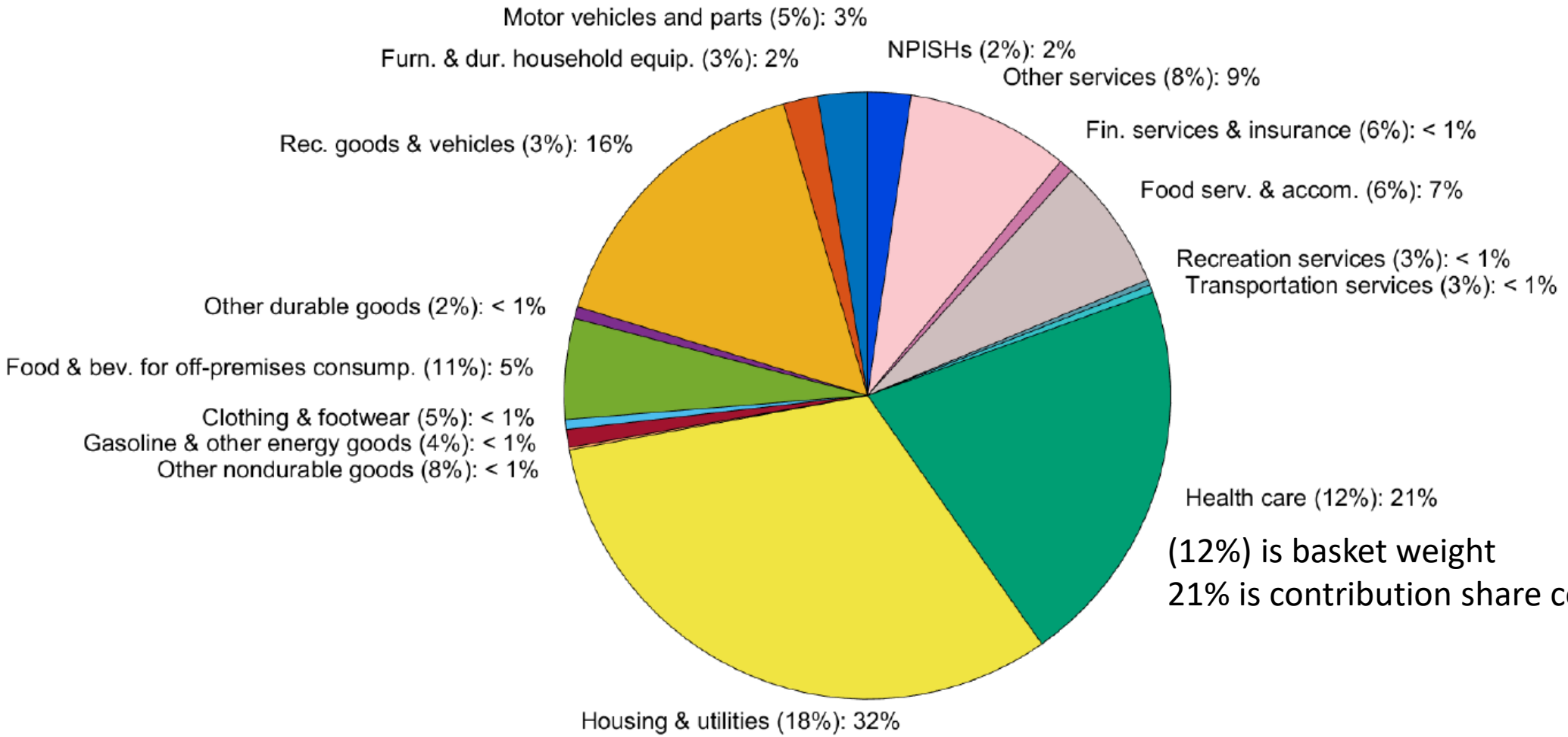
$$Share_i = \frac{\sum_{h=0}^{\infty} IRF^{sector}(h)}{\sum_{i=1}^{16} \sum_{h=0}^{\infty} IRF^{sector}(h)}$$

- In words,

$$\frac{\text{area under the curve of } \textbf{a} \text{ sector's IRFs}}{\text{area under the curve of } \textbf{all} \text{ sector's IRFs}}$$

- Initial impact of sector shock depends only on the basket weight, but the inflation persistence depends also on self persistence and spillover matrix

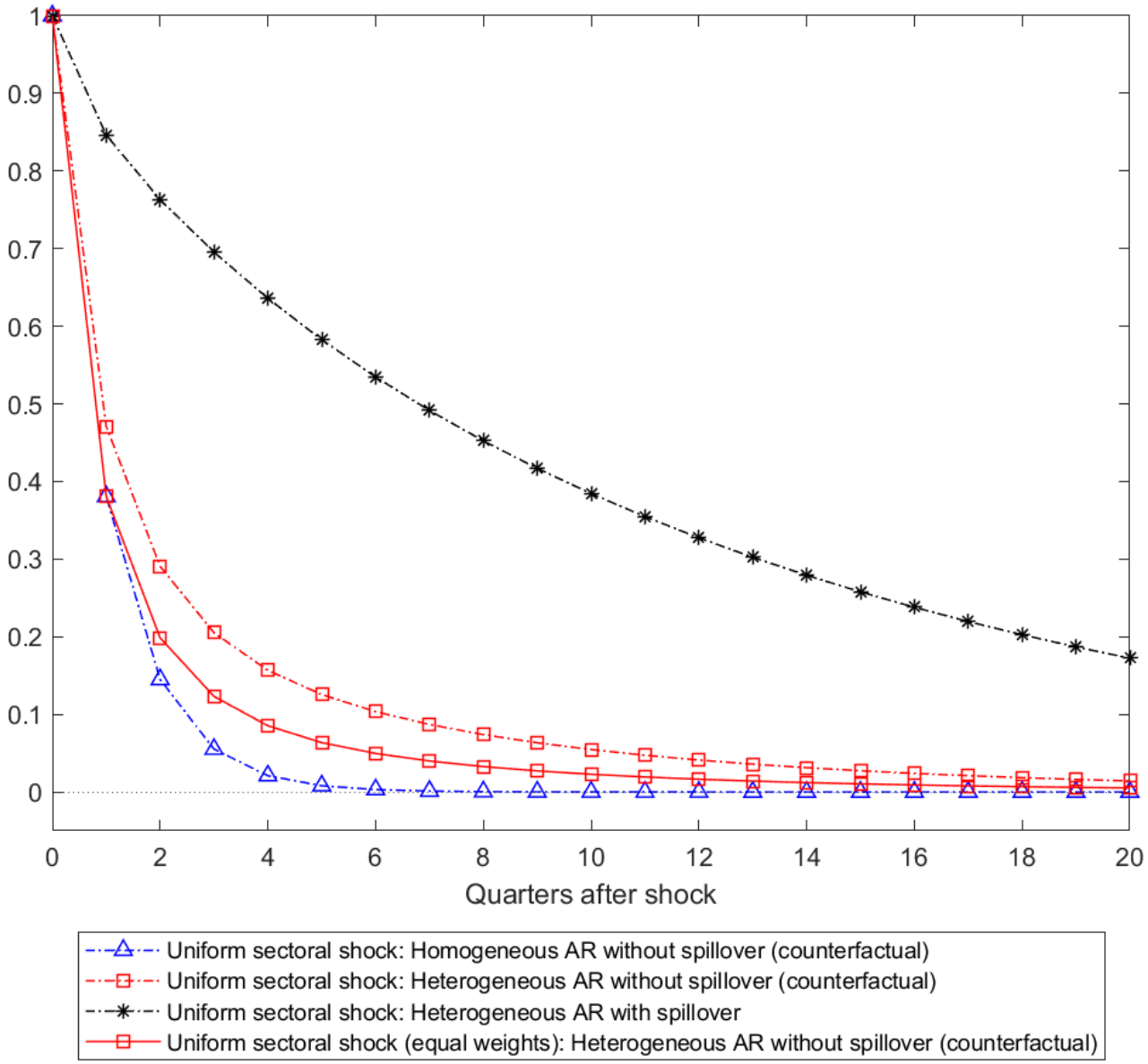
Sector granularity – some sectors are more important



(12%) is basket weight
21% is contribution share computed

- 3 sectors account 70% of inflation: housing, health care, recreational goods and vehicles, 80 % if add food services and food & beverage consumption

Sector heterogeneity vs spillover



- Blue = No heterogeneity at all
- Red = Heterogeneity but no spillover
- Black = everything.

- Takeaway

Heterogeneity increases persistence but sector spillover contribute much more

Roadmap

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- **Aggregate shock**
- Accounting for post 2020 inflation

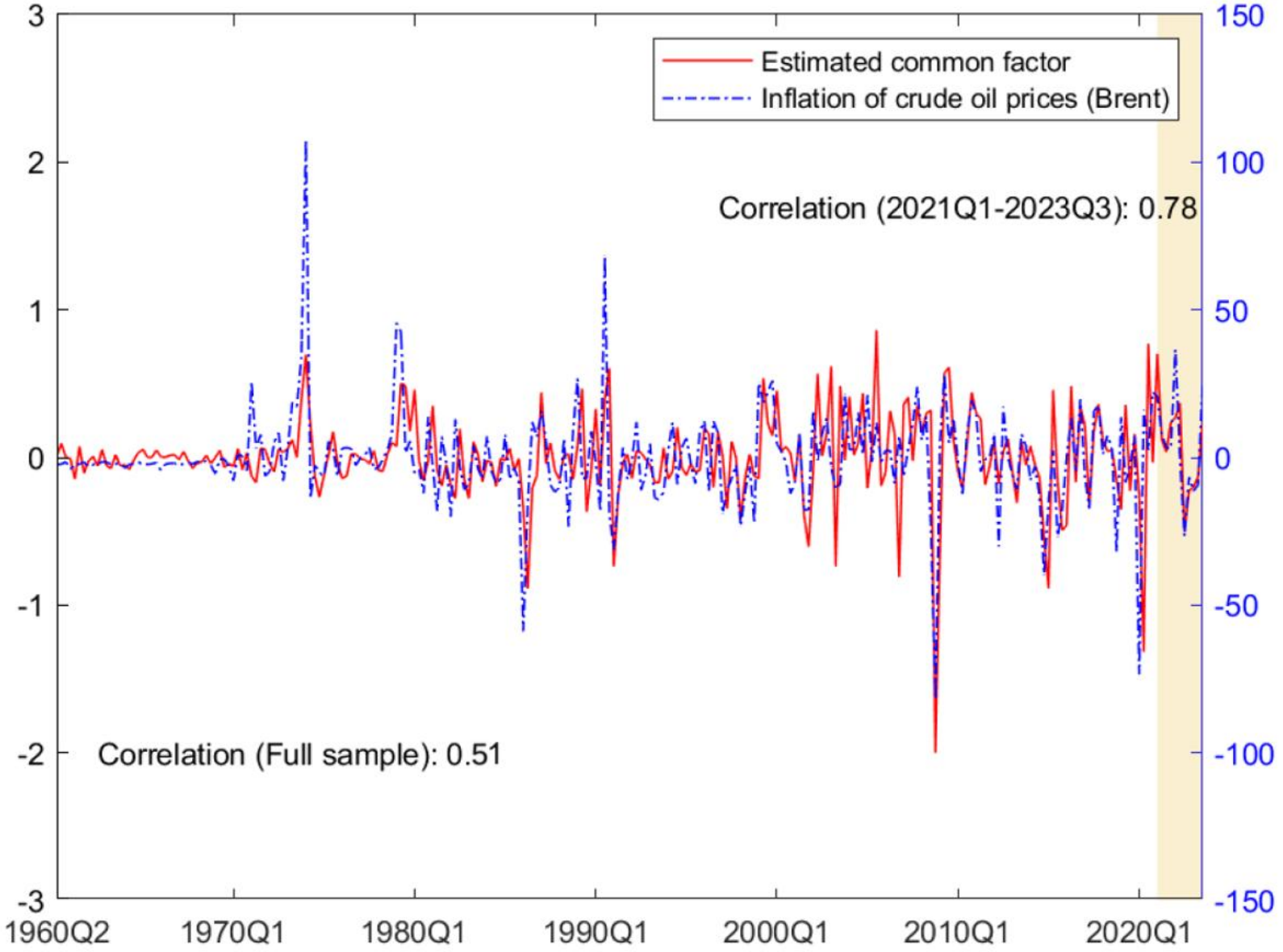
Aggregate shock

1. Estimated to have one aggregate factor

2. The aggregate factor has almost zero persistence (estimated 0.012)

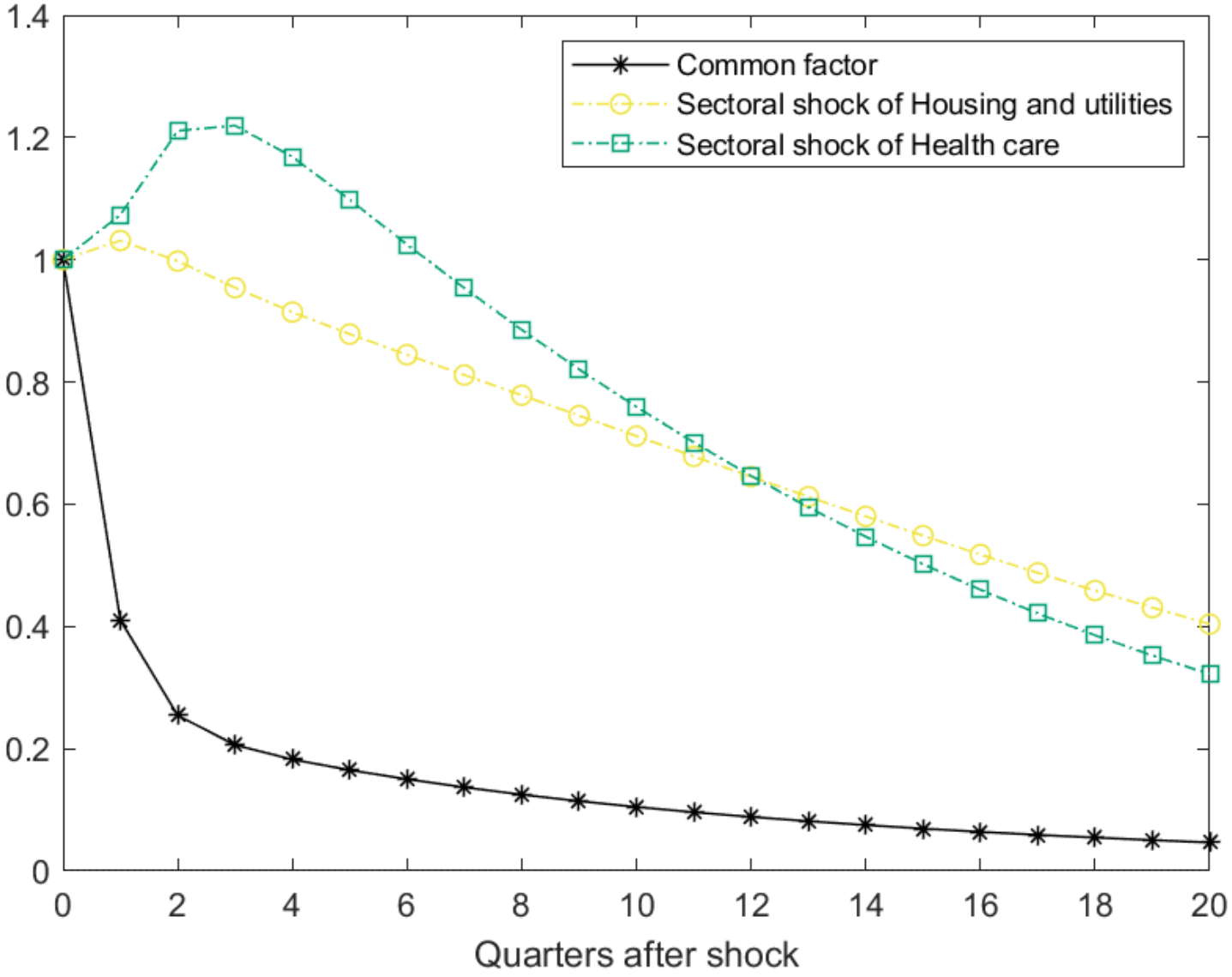
3. Aggregate factor looks very much like the crude oil price

Common shock and oil price



➔ To summarize, aggregate factor looks like change of oil price with low persistence

Summary

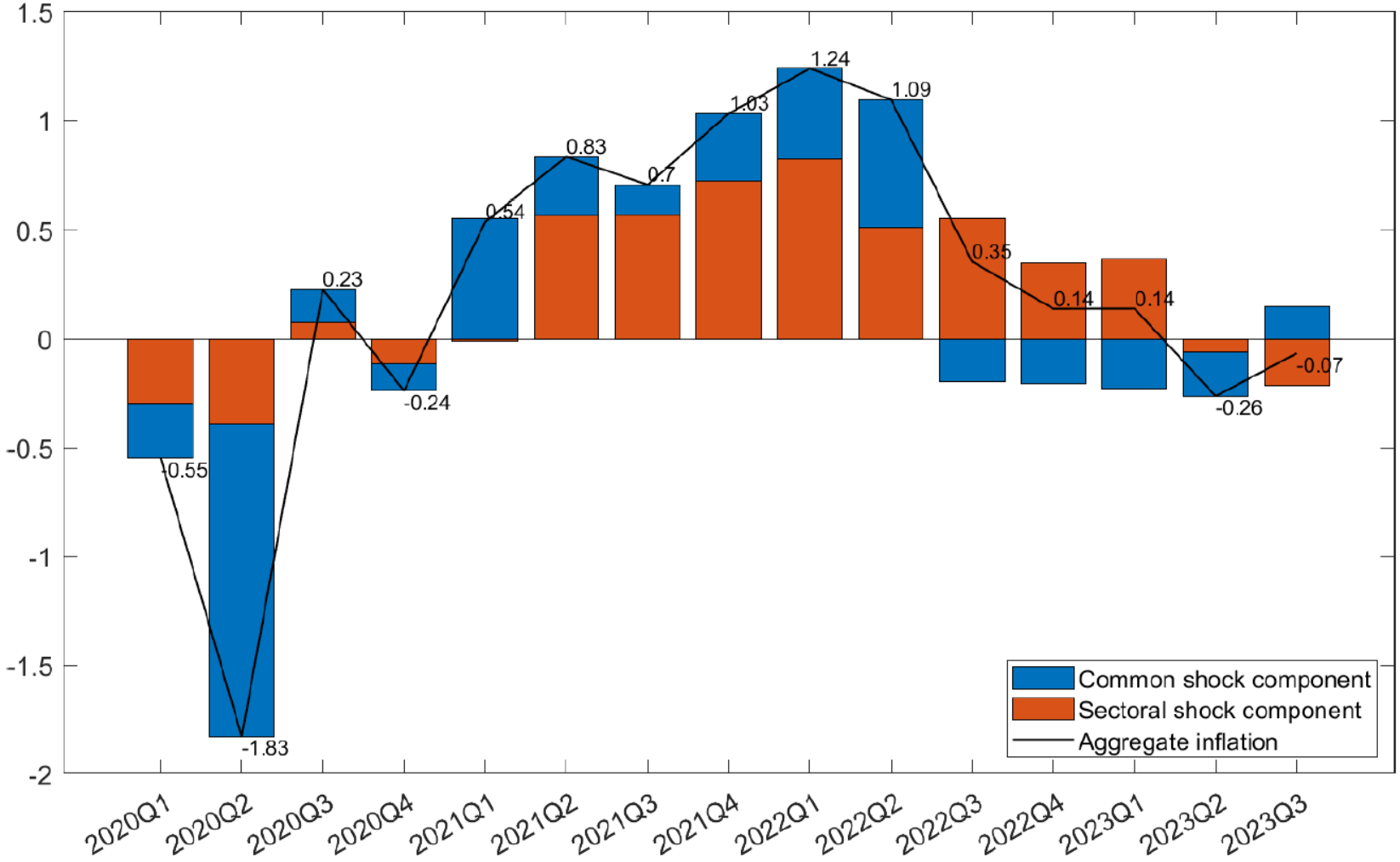


- Aggregate inflation is persistence or not depends on the relative contribution of aggregate shocks vs sector shocks

Roadmap

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- **Accounting for post 2020 inflation**

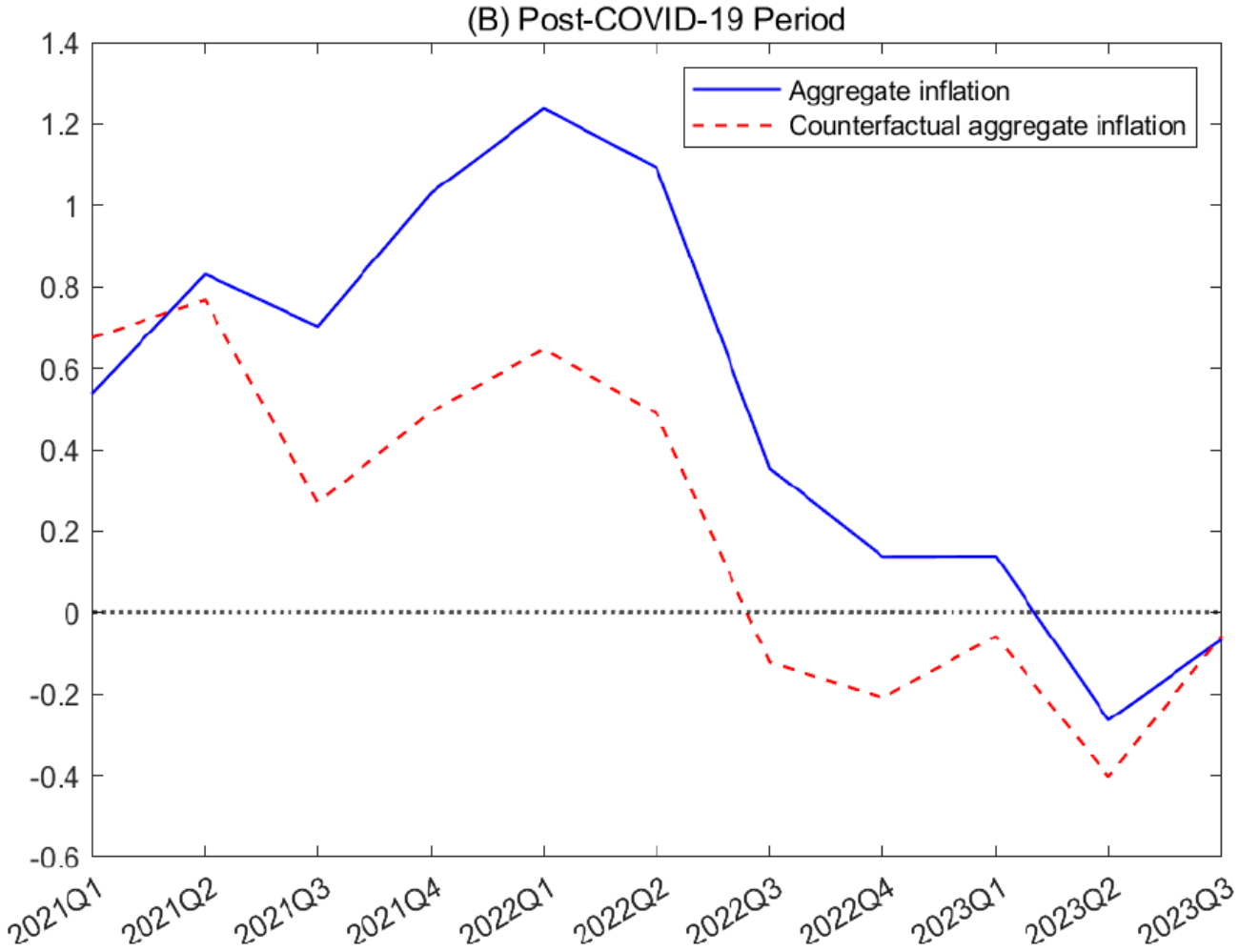
Post COVID-19 inflation decomposition



- Aggregate inflation variation explained by **common factor**

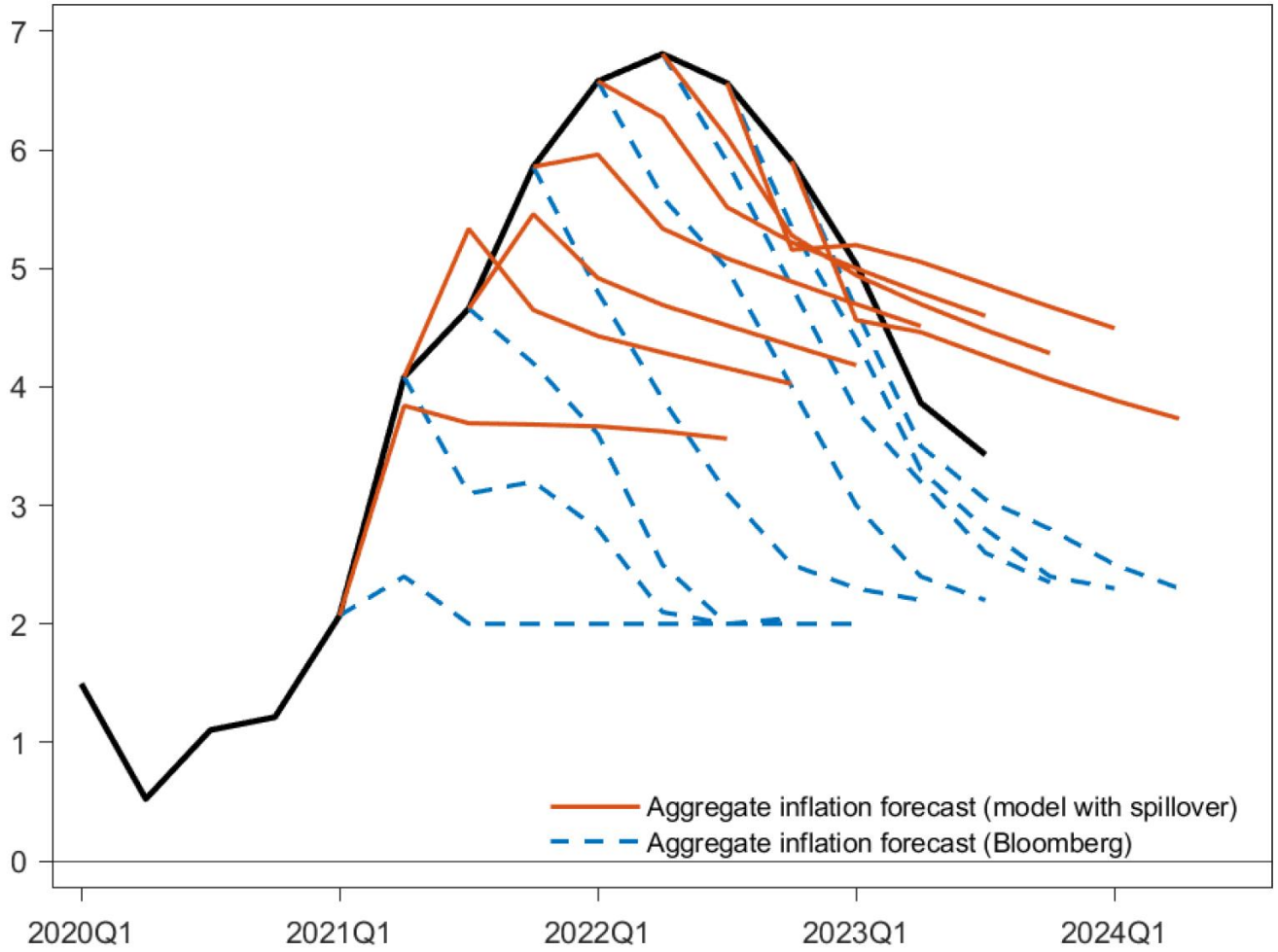
2010-2014 = 87%, 2015-2019 = 88%, 2021-2023 = 65%

Post COVID-19 inflation persistence

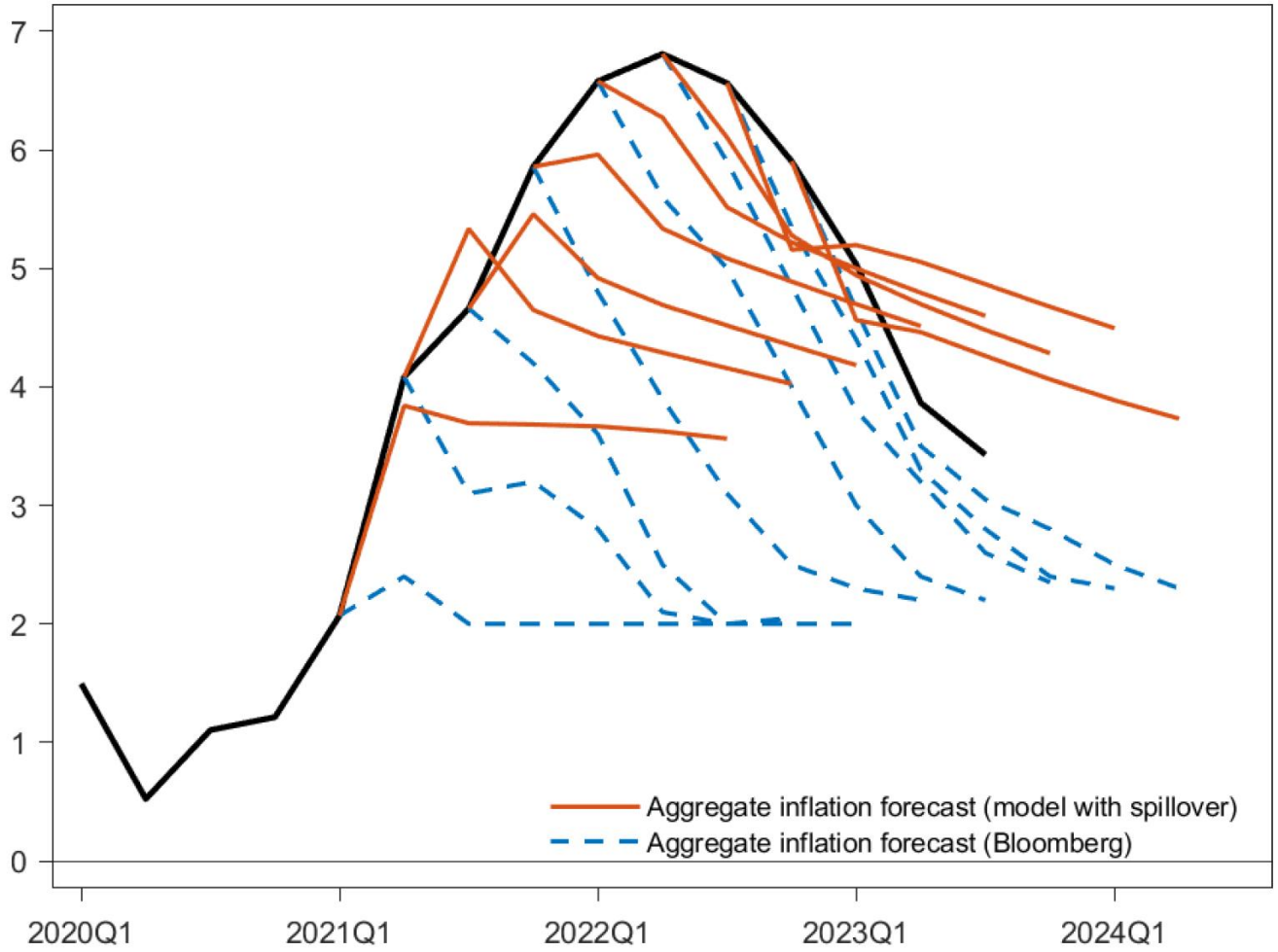


- Red line is the counterfactual spillover is assumed to be zero
- Overall price change will be 22% less, return to the mean 3 quarter earlier

Conclusion


7
6
5
4
3
2
1
0
2020Q1
2021Q1
2022Q1
2023Q1
2024Q1
Aggregate inflation forecast (model with spillover)
Aggregate inflation forecast (Bloomberg)

Conclusion



- A simple but flexible framework to study sector network spillover and the implications to aggregate inflation persistence
- Rich sectoral spillover and it tend to increase inflation persistence
- Sectoral shock is more important in post 2020s high inflation period

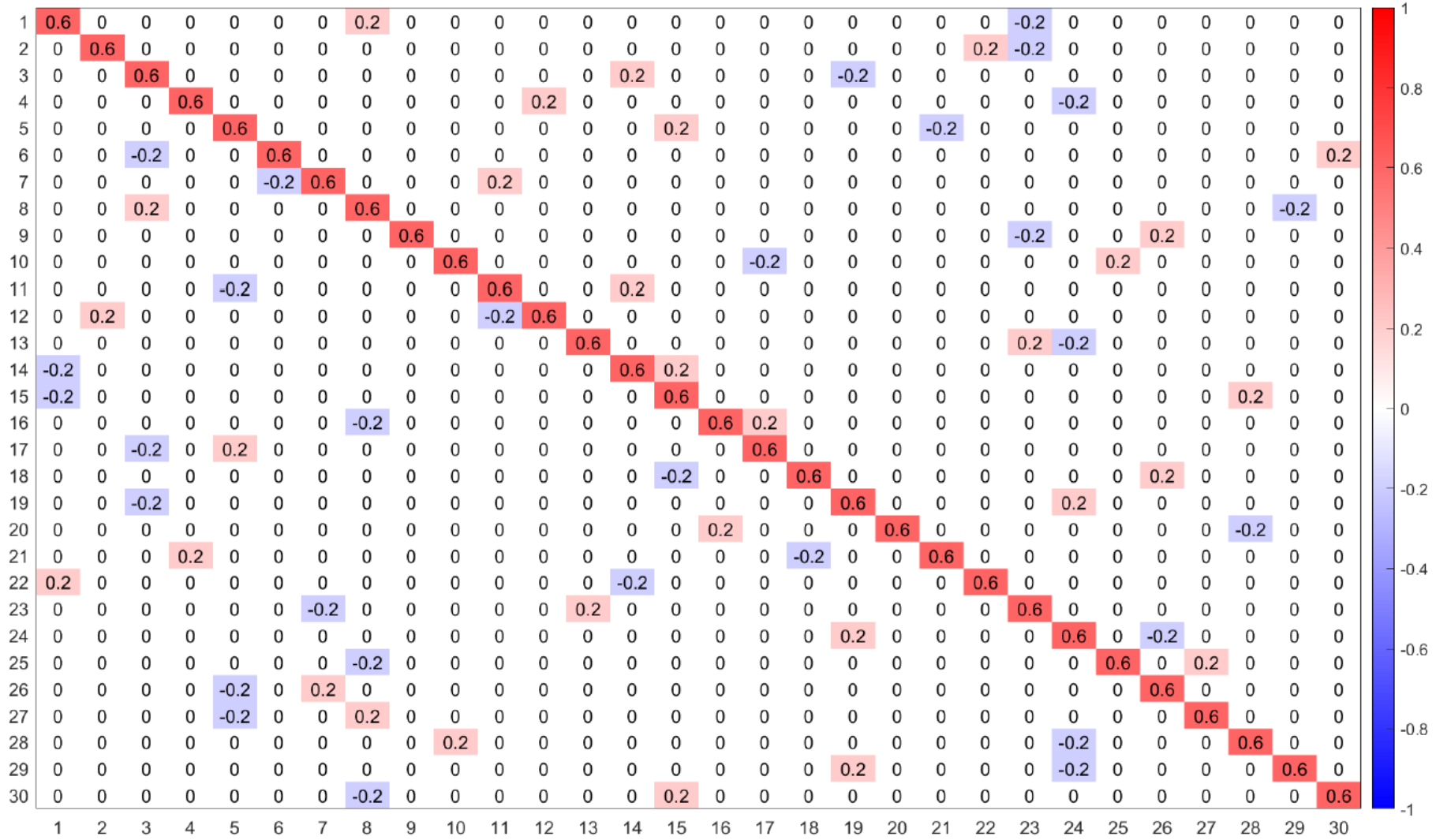
Conclusion

- A simple but flexible framework to study sector network spillover and the implications to aggregate inflation persistence
- Rich sectoral spillover and it tend to increase inflation persistence
- Sectoral shock is more important in post 2020s high inflation period
- ➔relatively persistent inflation

THANK YOU

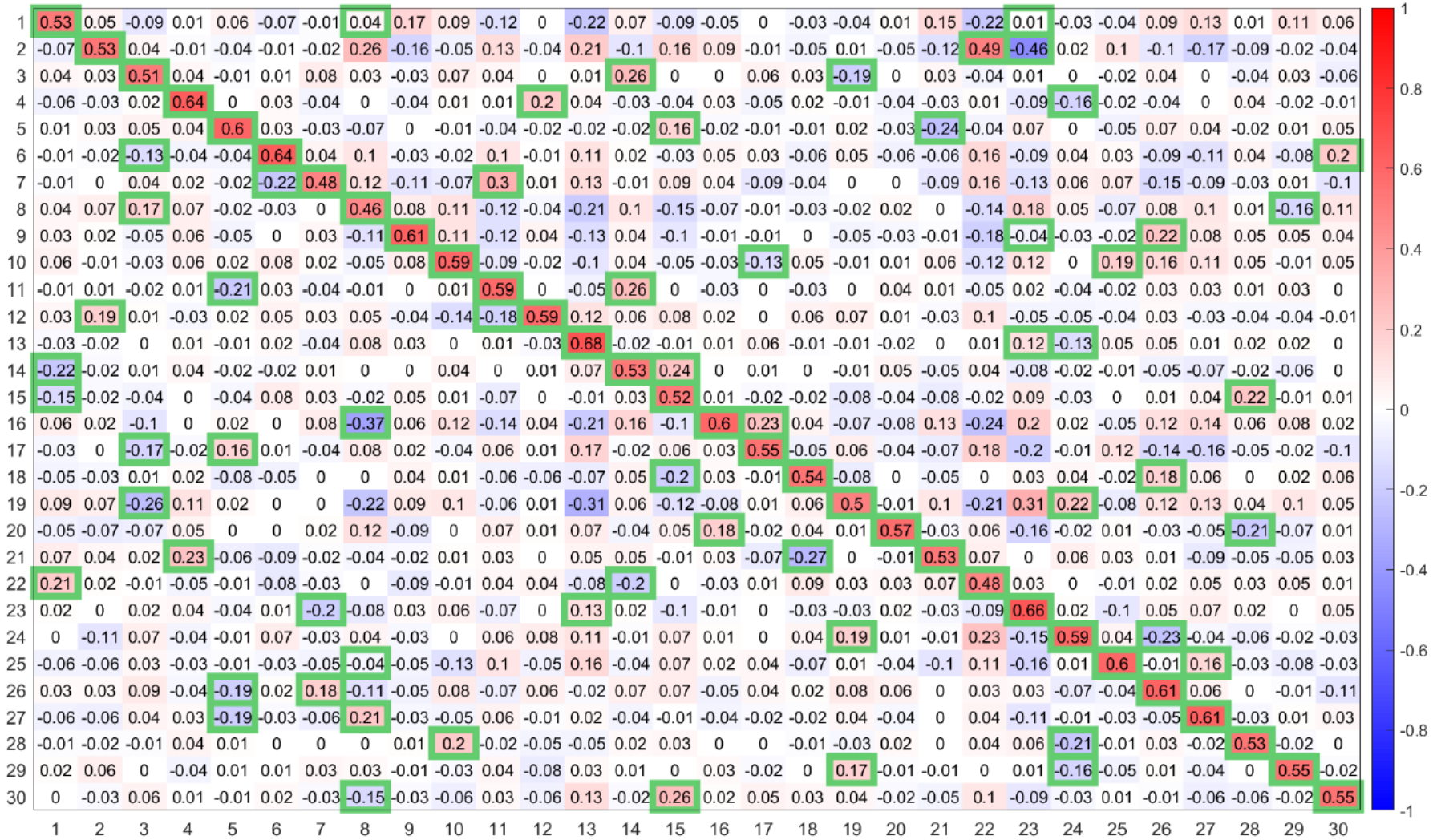
Why Lasso – simulation with N=30, T=500

Figure: True Autoregressive Matrix A



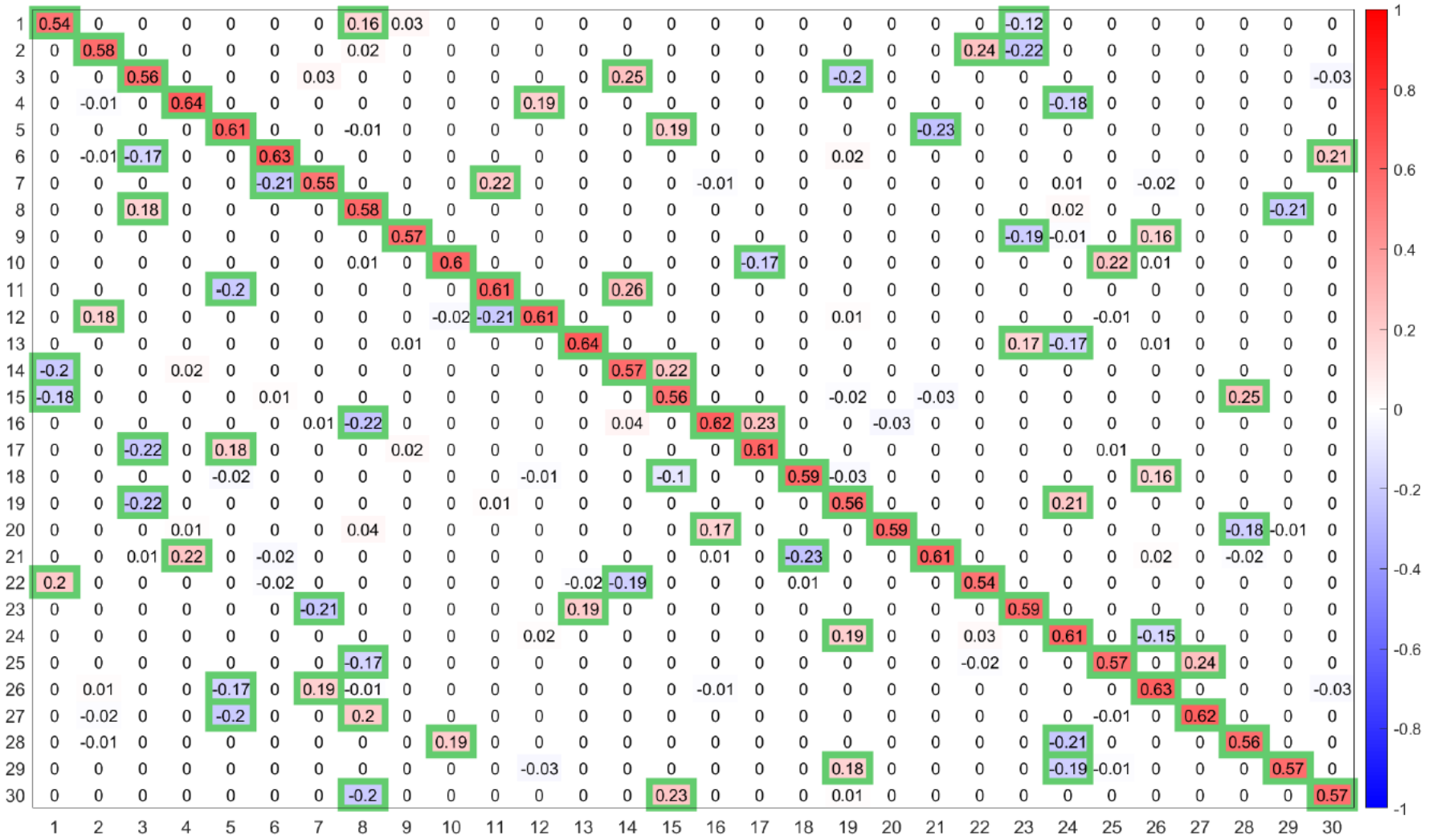
Why Lasso – simulation with N=30, T=500

Figure: Estimated Autoregressive Matrix without Lasso

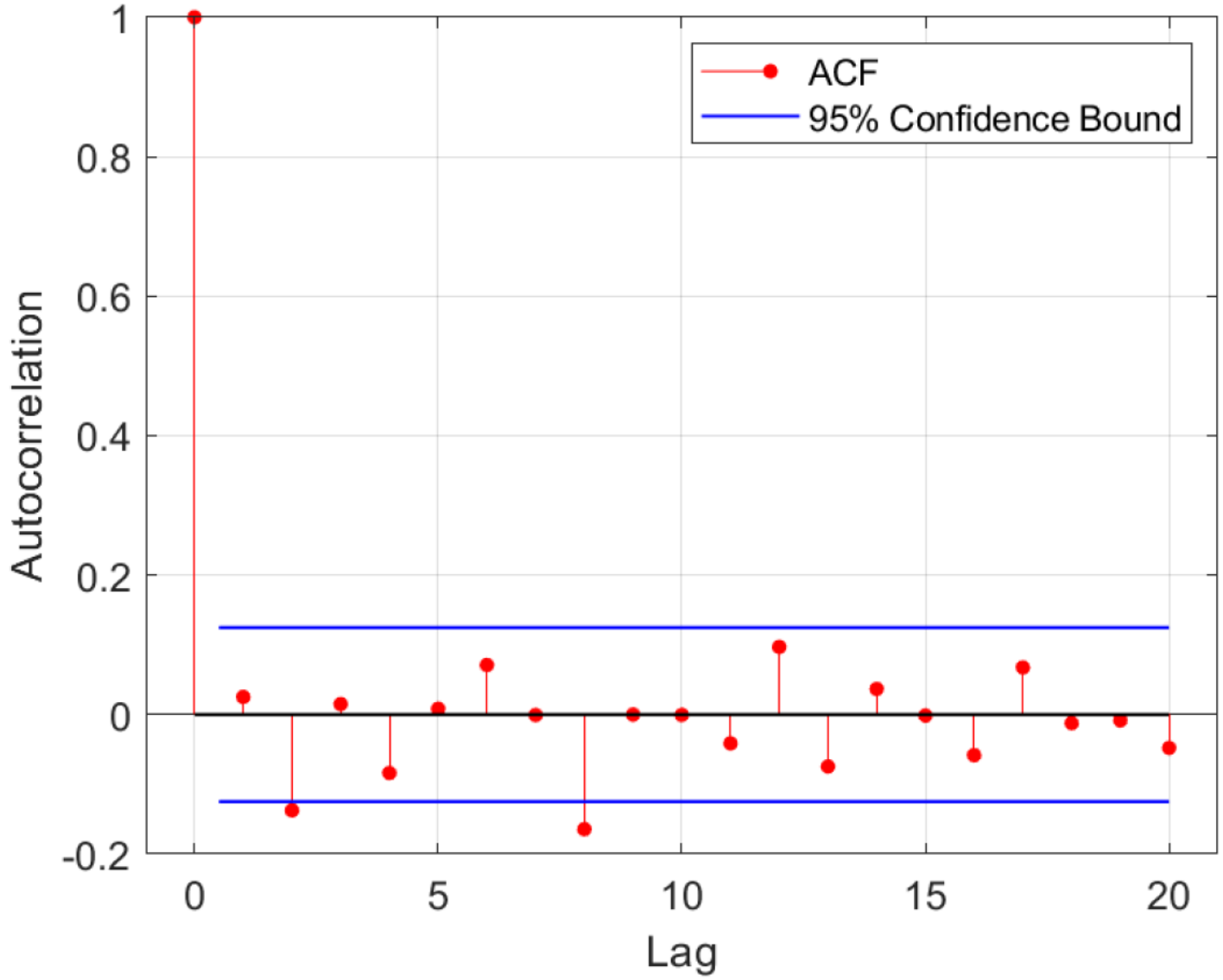


Why Lasso – simulation with N=30, T=500

Figure: Estimated Autoregressive Matrix with Lasso



AR0 for common factor



Factor Loading

